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Extended Elastic Impedance Inversion in Tight Sand Reservoirs

O. Tuwayrib, S. Misra, and J. Felemban, URE&CD Saudi Aramco, Saudi Arabia

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Surface seismic data plays a pivotal role in delineating hydrocarbon reservoirs and understanding the target reservoir properties. It has proven to be a valuable tool to identify sweet spots and optimize drilling practices and placements for horizontal wells (Treadgold et al., 2011). Tight sand reservoirs have increasingly become a prime target for oil and gas industry. The unconventionally low porosities encountered in such reservoirs are often a challenge for surface seismic data due to the limitations in the seismic response. Hence, the delineation and characterization of the reservoir porosity is hindered when using the conventional poststack inversion methods.

In 1999, Connolly introduced the idea of Elastic Impedance (EI) as a generalization of Acoustic Impedance (AI) with varying incidence angles. The work was then expanded to the Extended Elastic Impedance (EEI) for reservoir properties predictions (Whitcombe et al., 2000). EEI attributes have proven very valuable for unconventional targets to improve reservoir predictions accuracy where minimal seismic responses are detected from surface seismic data. Sorting through different angles to capture the optimal response for the desired property can enhance the prediction accuracy by a considerable margin.

Elastic impedance is believed to be more beneficial in reservoir characterization overall, not only in predicting the reservoir properties, but also in mapping the reservoir extent as well as the top and bottom of target reservoirs. The ability to add additional information from the shear impedance to the estimated acoustic impedance provides valuable information to differentiate lithology and fluids (Ogagarue, 2016)

Introduction

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Poststack inversion is often sufficient for porosity predictions in conventional reservoirs, whereas simultaneous prestack inversion has proven to be a more robust approach to estimate reservoir properties. By moving to the EEI domain, we expect an increase in accuracy for porosity estimations. This paper aims to highlight the difference between conventional Poststack porosity predictions compared to the EEI approach in unconventional tight sand reservoir.

Geological Background

The target formation forms a very unique siliciclastic-dominated succession that was influenced by a very large ice sheet that covered most of the paleo-continent of Gondwana, the sedimentary record of this glacial event has been documented from many parts of northern Africa, eastern and northern side of the Kingdom of Saudi Arabia (KSA) (Fig.1).

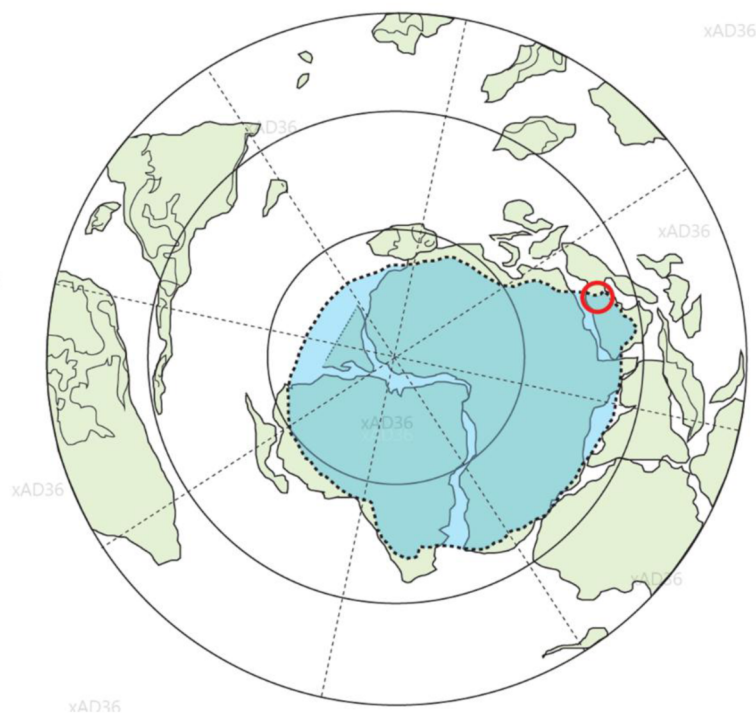


Figure 1—Paleogeographic reconstruction of the continent of Gondwana showing the extent of the ice sheets. (Rees et al., 2016)

In the study area, the target formation forms a thick pile of massive, amalgamated sandstone beds that are interbedded with very thin discontinuous layers of silty sandstone lithologies. Deposition of the sandstones occurred mainly by glacio-marine turbidite fans which may influence lateral rock property variability in different turbidite fans. This type of reservoir quality variation may also influence lateral stratigraphic sealing, and potentially lateral stratigraphic trapping (Fig.2).

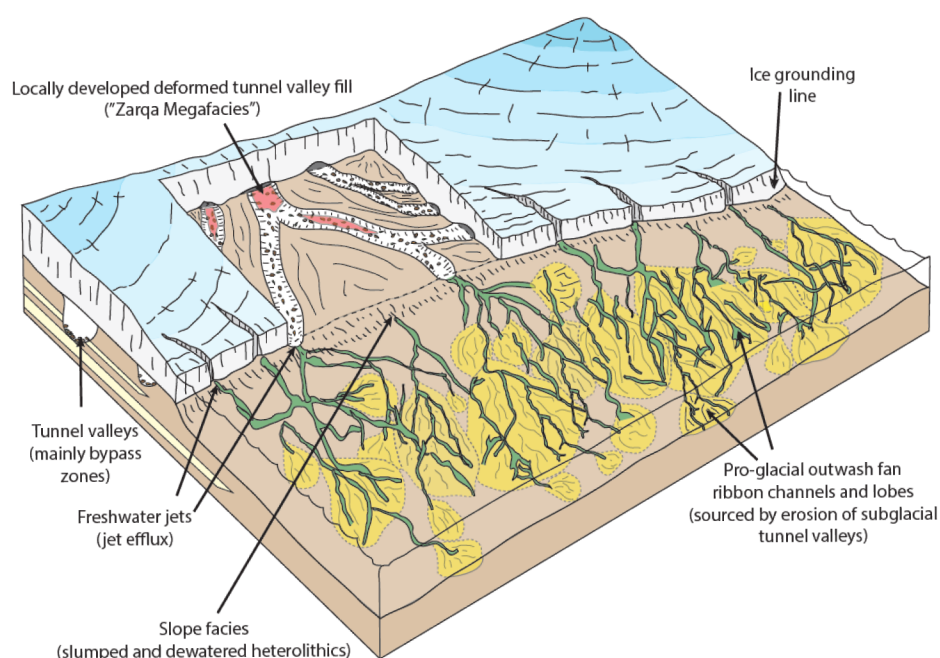


Figure 2—A simplified conceptual depositional model showing the general glacio-genic depositional settings from sub-glacial to proglacial (or glacio-marine) environment. Different fan lobes represent the lateral heterogeneity. (Rees et al., 2015)

Generally, the target formation is a gas-to-condensate-bearing zone with a high lateral hydrocarbon variability. This hydrocarbon variability is mostly attributed to the lateral reservoir quality which may accommodate condensates-rich zones in some areas, and gas-rich zones in other areas.

The structural elements are also crucial in forming the hydrocarbon trapping mechanism. The study area has undergone a series of tectonic events, the Hercynian tectonic events believed to be the major one which resulted in most of the structural traps. Both structural and stratigraphic traps are believed to be formed prior to the beginning of the charge which is interpreted to be started at the time of early Triassic era.

It's critical to understand the two components of structural and stratigraphic trapping in order to define the hydrocarbon rich areas. Focusing only into one element may result into a failure of discovery as it has been experienced. Therefore, understanding the porosity distribution can be crucial in characterizing the reservoir extent and its potential stratigraphic trapping mechanisms.

Also, the recent hydrocarbon production data has revealed variations in fluid types along the same geological structure, which can be attributed to the lateral variability in the reservoir quality.

Seismic data can provide high quality insight into the reservoir complexity from the standpoint of structural understanding as well as the lateral variability. Ultimately, the seismic data can be used as a reliable sweet spotting tool to define the hydrocarbon rich areas within the target formation.

Methodology

Study Area & Data Preparation

The study area includes a total of eight well penetrations, evenly distributed in the entire study area. Two wells were excluded from the study due to poor log-quality. Later on, during the implementation of the EEI workflow, one more well was dropped from the analysis as it showed extremely low correlation with the target logs. Therefore, a total of 5 wells with a good overall distribution across the area were selected to feed into the study, out of which, one well was left as a blind test. The blind well was chosen because it is the farthest from the remaining wells, which will provide a good measure of the inversion predictability and reliability.

In order to achieve optimal results, it is crucial to check the data quality and processing workflows to ensure that the best-quality data are being utilized in the study. Petrophysical logs were quality-checked and conditioned to account for any spikes or missing log sections. The caliper log was primarily used as a direct indicator of hole conditions and to decide on the log intervals to be dismissed. Additionally, missing sections within the shear velocity logs (V_s) were interpolated using the relationship between compressional wave velocity (V_p) and V_s logs obtained from the remaining wells.

As the foundation for quantitative interpretation work, seismic data must be carefully handled in the processing stage to avoid altering the relative amplitude response. The 3D seismic data utilized for the study was recently reprocessed to preserve the relative amplitude. The angle stacks were generated with central angles ranging from 5 to 53 degrees.

However, for the EEI implementation, a maximum angle of 45 degrees was used. The inversion interval was carefully chosen to include the reservoir top and bottom, in addition to a buffer zone in order to keep the wavelet stationarity in mind.

AVO Analysis

The partial angle stacks were sorted to form pseudo-gathers for Amplitude Versus Offset (AVO) analysis. AVO forward modeling was carried out to determine the optimal angle range for the study as well as to ascertain the AVO feasibility. Well synthetics were generated using Zoeppritz equation. Intercept and gradient analysis of the pseudo gathers compared to the well synthetics showed a good data quality up to 45 degrees, where we observed consistent gradient and intercept when compared with the AVO synthetics. (Fig.3).

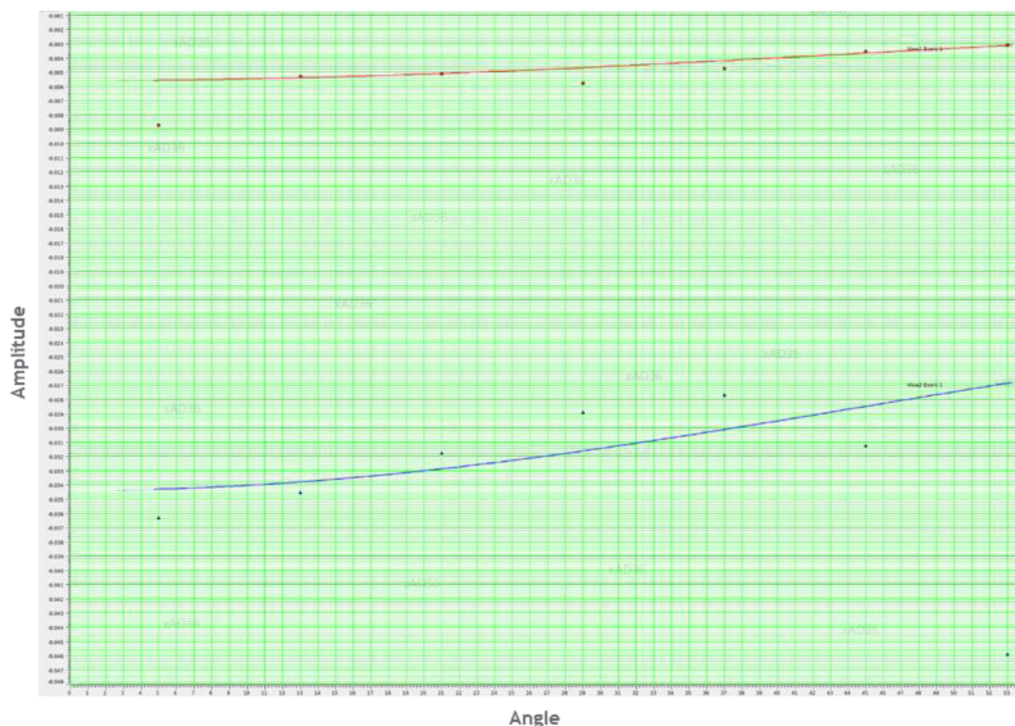


Figure 3—AVO analysis for gradient and intercept of well synthetic (Blue) in comparison to the generated pseudo gathers (Red) from the angle stacks. Negative intercept and positive gradient consistent with the AVO synthetics are observed.

Two AVO attribute volumes, intercept and gradient, were then obtained from the pseudo gathers within the specified angle range.

Wavelet Extraction and Well Ties

Statistical wavelets were extracted for each well within ± 50 inline and crossline range from the well location within an interval of 800ms in order to ensure wavelet stationarity. A constant phase wavelet with a 200ms length was found to be sufficient with minimum noise for well ties. The well-tie accuracy ranged from 50 to 70%. Wells with poor ties (less than 50%) were dismissed from the study.

Acoustic Impedence (AI) & Gradient impedance (GI) Logs

Computation of AI and GI logs is necessary prior to building the corresponding low frequency models. AI logs are straight-forward calculations from Vp and density logs. GI logs however, require the normalization factors for Vp, Vs, and density. This is critical to account for the silty/shaly layers within the reservoir and the GI values departing from such layers. The normalization constants were obtained from the Vp, Vs, and density logs within the target interval.

The GI log is calculated using the following equation:

$$GI = \left(\frac{Vp0}{Rho0} \right) \times \left(\frac{Vp}{Vp0} \right)^{\cos(chi) + \sin(chi)} \times \left(\frac{Rho}{Rho0} \right)^{\cos(chi) - 4 \times K \times \sin(chi)} \times \left(\frac{Vs}{Vs0} \right)^{-8 \times K \times \sin(chi)} \quad (1)$$

Where:

chi = 90

Vp0, Vs0, and Rho0: Compressional wave velocity, shear wave velocity, and density normalization constants

Vp, Vs, and Rho: compressional wave velocity, shear wave velocity, and density

$$K = \left(\frac{Vs0}{Vp0} \right)^2$$

Once the GI log is computed, the pseudo density corresponding to the gradient impedance must be calculated by dividing GI over Vp.

Poststack Inversion

AI and GI attributes are generated through inversion prior to moving to the extended elastic impedance domain. As it is well known, seismic data is deprived of low frequency information, this is why it is necessary to accommodate for this data with an initial model. The corresponding low frequency model for each attribute was built utilizing Vp, AI, GI, density, and pseudo density logs. Two seismic horizons were incorporated into the model as well as the inversion, which are sufficient for the target interval. The model was built to a frequency of up to 15 Hz according to the spectrum analysis of the seismic data.

Overall, we see a better correlation in the AI low frequency model prediction with the blind well compared to GI initial model (Fig.4).

This can be attributed to the uncertainties in the pseudo density calculations as well as the shear wave velocity logs. However, the signatures encountered in the GI initial model are believed to be more representative of the different facies encountered in the area.

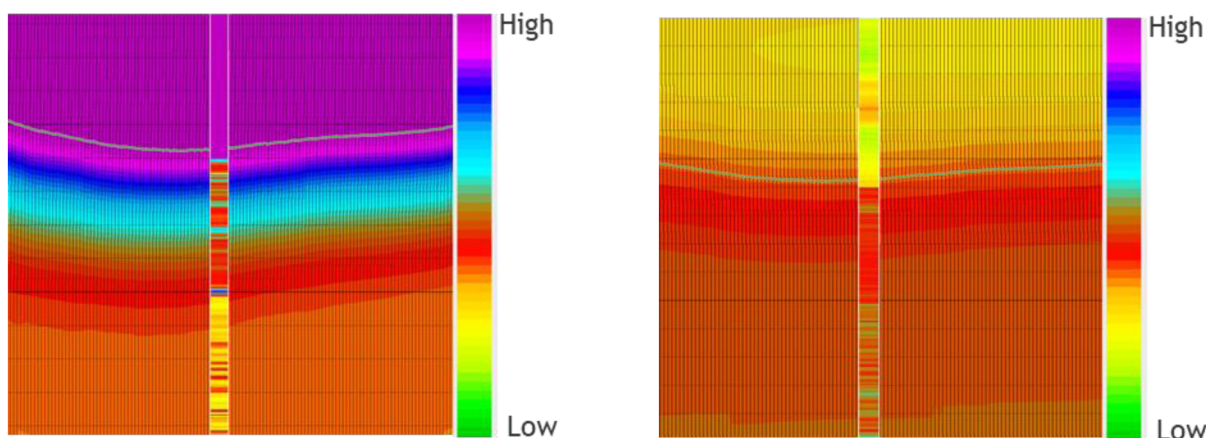


Figure 4—Comparison with the blind well log of the Low frequency model results for Acoustic Impedance (Right) and Gradient Impedance (Left).

Using the corresponding low frequency models, AI and GI inverted volumes were generated with a correlation of 95% and 92%, respectively (Fig.5).

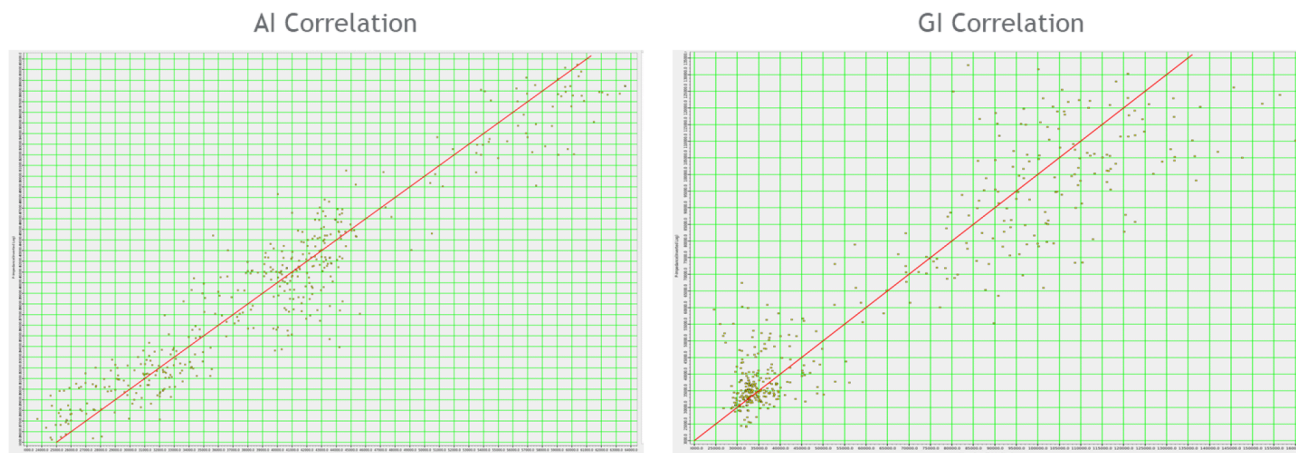


Figure 5—Correlation with well logs for Acoustic Impedance (AI) and Gradient Impedance (GI).

The Poststack porosity prediction can now be estimated using the AI inverted volume. From well logs, AI and porosity were cross plotted to obtain the porosity transform. The transform was then applied to produce the inverted porosity volume.

Angle Rotations

Poststack inverted porosities are obtained for the normal incidence angle zero. EEI enhances the correlation by scanning through different rotation angles, also known as the chi-angle, and identifying the angle that maximized the correlation with porosity. Utilizing AI, GI, and porosity logs, we computed the correlation for each chi-angle at the well locations for the detrended logs. The correlation between AI & GI with porosity is then plotted to find the Chi angle with maximum correlation (Fig.6).

As porosity is inversely related to AI, we expect chi-angle to be close to -180° for maximum correlation, which is confirmed in figure 6, where the maximum correlation chi-angle is found to be around -165 degrees. For comparison, angle rotation was also conducted to investigate the maximum correlation with other reservoir properties, such as water saturation and Gamma Ray signature.

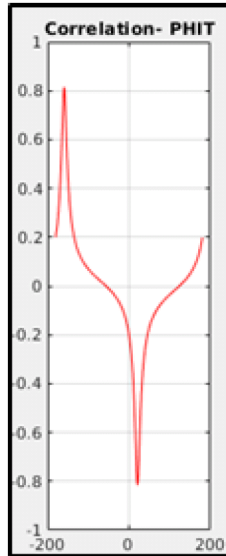


Figure 6—EEI for different Chi-angles vs. correlation coefficient.
Maximum correlation of 0.86 was observed for a chi-angle of -165°.

Two wells out of the 4 wells used in the study displayed relatively low maximum correlation with Chi angle, despite no issues with the AI and GI logs. After a closer look at the porosity distribution in these two wells, the porosity readings from the well logs were found to be either extremely tight or abnormal when compared with the other two wells. The correlation of the wells with AI alone was found to be less than 30%, while the correlation with EEI was more than 45%. This clearly highlights the challenges of predicting porosity in tight sand reservoirs, and the value addition of the EEI workflow in predicting porosity distribution more accurately.

Extended Elastic Impedance (EEI)

Once the maximum correlation angle between AI and GI with porosity was obtained, the angle was used to generate EEI logs for all wells. The logs generated represent the EEI response at that specific Chi angle of -165°. The EEI logs were calculated using the below equation:

$$EEI(chi) = AI0 \times \left(\frac{AI}{AI0} \right)^{\cos(chi)} \times \left(\frac{GI}{AI0} \right)^{\sin(chi)} \quad (2)$$

Where:

Chi = Chi angle for porosity

AI0= Vp0 * Rho0

AI = Acoustic Impedance

GI = Gradient Impedance

Vp0, Vs0, and Rho0: Compressional wave velocity, shear wave velocity, and density normalization constants.

Vp, Vs, and Rho: compressional wave velocity, shear wave velocity, and density

Using the same relationship, the EEI volume was calculated for the given angle. The volume is generated using the previously inverted AI and GI attributes.

In order to obtain porosity, the computed EEI logs were cross-plotted with the porosity logs and the necessary transform equation was established to convert the EEI volume to the porosity domain. The correlation between EEI and porosity was found to be reliable with 74% correlation. Similarly, for the Poststack inversion, the correlation between AI and porosity was only up to 66% (Fig.7).

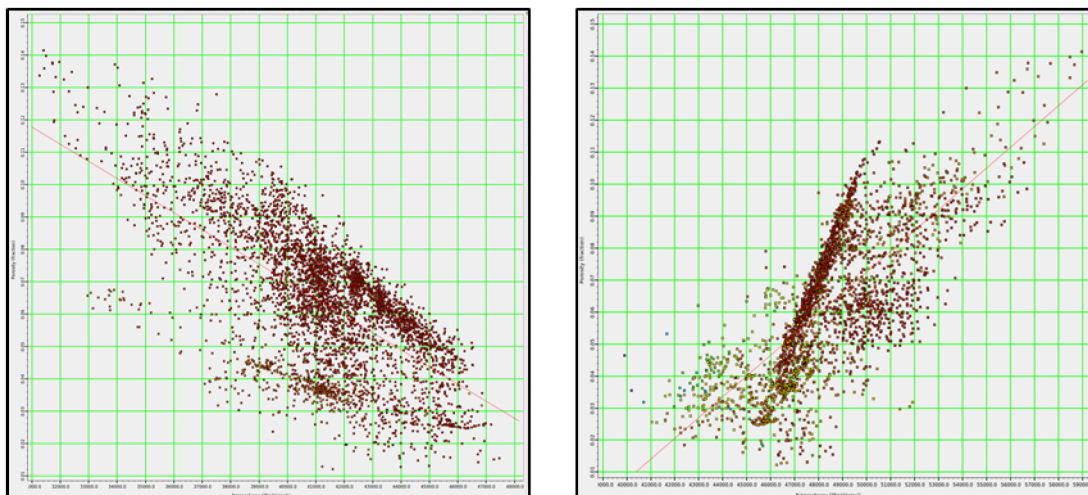


Figure 7—Cross plot showing the obtained correlation with porosity from Acoustic Impedance (left) and Extended Elastic Impedance (Right). The correlation coefficient is 66% and 74% from left to right.

On the well logs scale, we observe the enhancement in prediction accuracy as the correlation between EEI vs Porosity increased to 74% from 66% in the AI vs Porosity. In addition, we see a more compact cluster in the EEI domain compared to AI, increasing the reliability of the transform equation.

As discussed earlier, tight sand reservoirs are challenging due to their very low porosities, which hinder the seismic response significantly compared to the conventional targets. In addition to the inherent challenges with characterizing tight sand reservoirs, the formation is relatively deep, where we expect less high frequency components due to the high frequency attenuation with depth. Such challenges must be taken into considerations when assessing the correlation between impedance and reservoir properties.

Results

We assessed the results of the inverted AI and GI volumes, as they are inputs for porosity predictions. Both inverted attributes displayed good correlation with blind wells (Fig.8).

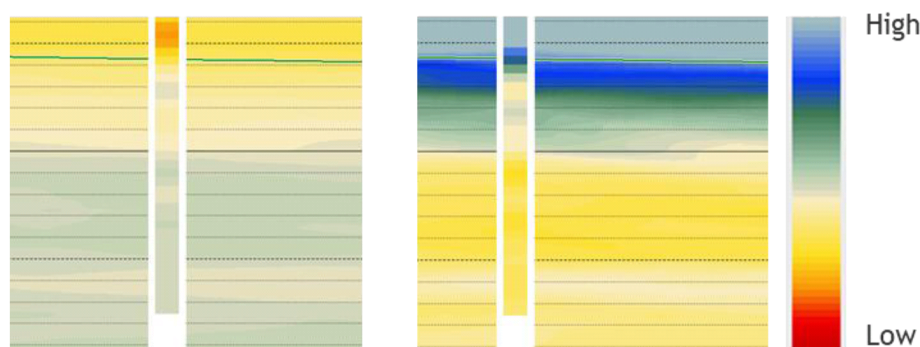


Figure 8—Inverted Acoustic Impedance (Left) and inverted Gradient Impedance (Right) compared to the blind well log. Well logs were filtered to the seismic frequency in this display.

In the GI volume we can observe a good representation of the interface between the target reservoir and the overlain layer, where the top reservoir zone is usually characterized with a lower porosity zone. However, this is not detected on AI, where a smoother transition between the two layers is seen. This implies that incorporating the gradient impedance and changes with offset can be beneficial in detecting certain features that could otherwise be overseen in zero incidence data. The angle rotations allow for better separation between porosity values in EEI domain when compared to the separation observed on AI alone.

By implementing the established relationship between EEI and porosity, and similarly with AI, we obtained two separate inverted porosity volumes. The volumes are displayed against the blind well to assess the prediction accuracy of porosity (Fig.9).

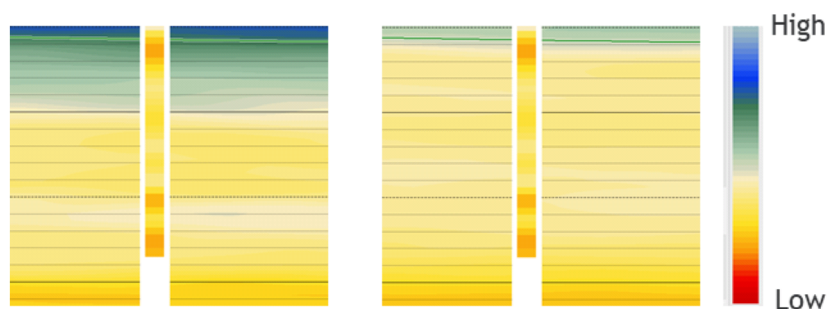


Figure 9—Inverted porosity from Acoustic Impedance (Left) and inverted porosity from Extended Elastic Impedance (Right) compare to the blind well log. Well logs were filtered to the seismic frequency in this display.

The well log was filtered to the seismic frequencies for a fair assessment of prediction accuracy and to account for the difference in resolution. We can see an overall better correlation between the porosity log from the well with EEI generated porosity compared to AI prediction. Poststack porosity volume fails to capture the immediate transition between the top of the target reservoir and the overlaying layer, where we see a continuation of the high porosities into the target reservoir. In contrast, EEI predictions seem to match very well with the blind test log and better captures the transition between the two layers.

In addition to the significant overestimation of the top layer in the AI based porosities, the Poststack predictions have produced spurious high porosity zones within the bottom half of the reservoir. This can be extremely misleading when looking for porosity sweet spots to place lateral wells within the tight sand reservoirs. EEI based predictions seem to match very well with the blind test across the target interval, with no significant over or underestimations for porosity.

Lateral variations within tight reservoirs are minimal, therefore, we do not expect to see substantial changes in porosity away from the wells. EEI based porosities accurately captures this consistency and subtle lateral variations we expect to see in the area. On the other hand, Poststack porosities produced multiple intervals of sudden lateral increase in reservoir porosities. This overestimation can be seen on the AI vs Porosity plot displayed in figure 7. At low porosity values, the transform seems to overestimate porosities due to the wide distribution across the AI axis corresponding to low porosities.

Conclusions

Pre-stack EEI inversion allows for improved predictions of porosity in tight unconventional reservoir settings. The EEI-based approach enhances the predictability when compared to conventional post-stack inversion workflow. The porosity predictions from the pre-stack EEI domain provide a valuable insight to determine reservoir sweet spots and optimize lateral well placement.

Upon applying both approaches to the blind well, the EEI-based approach showed a better correlation with the blind well in the EEI-based inverted porosity, in addition to an increase in overall correlation between EEI and porosity observed in the log domain. This implies that the elastic impedance provides valuable information that is otherwise undetected in the AI domain.

Prestack inversion is commonly believed to yield enhanced results when compared to Poststack porosity predictions in tight sand reservoirs. Furthermore, the enhancement in predictions accuracy is evident when incorporating the extended elastic impedance workflow into the analysis.

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